

# Srinivas Sivaratri

AI Engineer | LLM evaluation, model debugging, developer tools  
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srinivassivaratri.is-a.dev | Remote, India (IST, UTC+5:30) | Open to full-time and contract

## Summary

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AI engineer with an MS in Computer Science. Builds LLM systems and the evaluations that check them, and publishes the numbers, including the ones that lose to a baseline. Upstream work reviewed by better-auth and go-ethereum maintainers.

## Skills

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**AI and LLM:** LLM evaluation, RAG, hybrid retrieval (BM25, Reciprocal Rank Fusion, cross-encoder reranking), token-level tracing and replay, LoRA fine-tuning, diffusion models

**Machine Learning:** PyTorch, Hugging Face Transformers, PEFT, sentence-transformers

**Languages:** Python, TypeScript, Go, JavaScript, SQL, C

**Engineering:** FastAPI, Flask, Next.js, React, Node.js, Bun, PostgreSQL, Redis, SQLite, Docker, GitHub Actions, pytest, Vitest

## Experience

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**Software Engineer**, Independent | Remote May 2025 to present

- Traced an email-enumeration leak in better-auth to its Have I Been Pwned plugin and proposed a request-scoped skip flag with a regression test. Maintainers adopted the approach and shipped it (PR #5998).
- Designed "geth db truncate-freezer" for go-ethereum (PR #31351), revised across three core-maintainer review rounds to batch-process instead of holding about 15 GB in memory. The core team later built its own version.

**Software Engineer**, Accenture | Bangalore, India July 2021 to July 2022

- Wrote a Python ETL script that automated data processing, saved 50 labor hours per month, and cut errors.
- Implemented a GitHub Actions CI/CD pipeline.

## Projects

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**Bingo** | Python, NLTK, sentence-transformers, CLIP, Gemini [github.com/sivaratrisrinivas/Bingo](https://github.com/sivaratrisrinivas/Bingo)

- Built a command-line movie search over 5,000 films during Boot.dev's RAG course, with BM25 over an inverted index, chunked embeddings, Reciprocal Rank Fusion, cross-encoder reranking, image search, and LLM answers with citations.
- Built an 834-query known-item eval. On the test split, RRF with cross-encoder reranking put the target movie in the top 10 91.6% of the time against 86.3% for BM25, and plain RRF lost to BM25 alone. Rewrote BM25 to score from precomputed term statistics, cutting p95 latency from 3.4 s to 5 ms.

**SokoFlow** | Python, PyTorch, Flask [github.com/sivaratrisrinivas/sokoflow](https://github.com/sivaratrisrinivas/sokoflow)

- Trained a diffusion model (CNN board encoder, Transformer denoiser) that turns random moves into Sokoban plans, served on CPU through Flask and a Hugging Face Space.
- Solved 6.2% (15/240) of scrambled 8x8 boards against 94.2% for plain BFS search. Traced an earlier 20.8% figure to easy boards and published the correction.

**llmfr (LLM Flight Recorder)** | Python, PyTorch, Transformers, SQLite [github.com/sivaratrisrinivas/llm-flight-recorder](https://github.com/sivaratrisrinivas/llm-flight-recorder)

- Built a CLI that records a local LLM generation token by token, replays it, and names the first step where two runs split, tagging every later difference as fallout. The OpenAI adapter refuses to record when logprobs are missing.
- On pinned Qwen2.5-0.5B demos, it labels a seed-only change as a sampling split at token 0 and a temperature-only change as a decoding split at token 8. Compare takes 0.151 s median on CPU.

## Education

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**MS, Computer Science**, University of Alabama at Birmingham, AL, USA Aug 2023 to May 2025

**BS, Computer Science**, BV Raju Institute of Technology, India Aug 2017 to June 2021

**Boot.dev, coursework**, HTTP Servers in Go, RAG, CI/CD with GitHub Actions and Docker, Memory Management in C, SQL